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New report of *Melithaea delicata* Hickson, 1905 (subclass: octocorallia) from Little Andaman Island, India

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Melithaea octocoral (Family Melithaeidae) are represented by 15 species in Indian waters and 13 species were reported from Andaman and Nicobar Island. *Melithaea delicata* is reported herein as a new distributional record in Indian waters, from Little Andaman Island. This species is identified by morphological and sclerite structures and characterised by club-shaped sclerites of the calyx, pointed needle shape sclerites of the collar; spindles shaped in the calyx and coenenchyme. Updated distribution details of all 15 *Melithaea* species in Indian waters are provided.

[**Keywords:** Octocorals; Soft corals; Melithaeidae; *Melithaea delicata*; Little Andaman; Andaman and Nicobar Island; India]

Introduction

The subclass Octocorallia belongs to class Anthozoa, phylum Cnidaria and is commonly known as soft corals and seafans (Alcyonacea), blue corals (Helioporacea), sea pens, and sea pencils (Pennatulacea). Octocorallia is classified into 3 orders, 8 suborders, 55 families and about 3200 species^{1,2}. The genus *Melithaea* belongs to the order Alcyonacea, suborder Sleraxonia, family Melithaeidae and subfamily Melithaeinae. The family Melithaeidae is one of the 49 presently recognized families of octocorals^{2,3,4}. The genus *Melithaea* is distributed in the marine environment and is very common in warm and tropical waters from shallow to deep sea^{5,6,7,8}.

A total of 121 species of *Melithaea* have been reported by different researchers from different parts of the world (East Africa and Red Sea by Grasshoff⁹; Indian Ocean by Thomson¹⁰, Ofwegen^{11,12}, and Williams¹³ Indo-West Pacific by Ofwegen¹¹, Grasshoff¹⁴, Ofwegen, *et al.*¹⁵ Hoeksema and Ofwegen^{6,7}, and Matsumoto and Ofwegen¹⁶). The genus name *Melithaea* was placed on the official list of generic names and the node and internodes play a major role for morphological identification of the species¹⁴. The axis of Melithaeidae having short and long internodes, is short,

smooth, rod-shaped⁸. Recently the family Melithaeidae was recognized by Reijnen *et al.*,¹⁷ based on investigation of the DNA molecular phylogenetic relationship and synonymised several genera, namely, *Acabaria*, *Clathraria*, *Melithaea*, *Mopsella*, *Wrightella* coming under this family.

In India, an aggregate of 461 species of octocorals belong to 92 genera, 30 families, 8 suborders, and 3 orders. Of which, 301 species are from the east coast, 98 species from the west coast, 277 species from the Andaman and Nicobar Islands, 203 species from the Tamil Nadu, 40 species from the Lakshadweep Islands, 31 species from the Kerala coast, 29 species each from Andhra Pradesh and Orissa, 27 species from Pondicherry, 18 species from Gujarat, 15 species from Mumbai coast, 8 species from West Bengal, and 4 species from Karnataka¹⁸. Among them, 14 species are reported under the genus of *Melithaea*, while four species (*M. variabilis*, *M. philippinensis*, *M. ornate*, *M. pulchella*) are reported from Andaman sea¹⁹, three species (*M. andamanensis*, *M. biserialis*, *M. squamata*) from south-east coast of India²⁰, one species (*M. variabilis*) from Muttom Kerala coast²¹, and one species (*M. variabilis*) from Minicoy Atoll, Lakshadweep Islands²². Recently, 7 species (*M. braueri*, *M. caledonica*, *M. cinquemiglia*,

M. ochracea, *M. ouvea*, *M. rubeola*, *M. variabilis*) are reported under this genus from Andaman and Nicobar Islands^{23,24}. In this study, we present detailed description of the newly recorded species of *Melithaea delicata* from Little Andaman Island and updated list of species reported under the genus of *Melithaea* from Indian coast.

Materials and Methods

The samples were collected during the survey period from June 2013 to May 2016 by SCUBA diving in South Bay, Little Andaman and preserved in 70% ethanol²⁵. The specimens were identified based on the structural characteristics of the colony and sclerites structure. Sclerites were extracted by 5% sodium hypochlorite²⁶ and probed under the compound and stereo microscope (Labovision AXR 20; Leica–DFC 500) and underwater pictures were taken with the help of Canon G15 camera. The identified samples were deposited in the National Zoological Collection of ZSI, Port Blair.

Results

In India, a total of 15 species belonging to the genus *Melithaea*, family Melithaeidae, 13 species of *Melithaea* were reported from Andaman and Nicobar Islands. Among them, one species *Melithaea delicata* was identified as the new distribution record to Indian waters from the Little Andaman Island on the basis of taxonomical attributes. The taxonomical identification was made based on the morphological variations, sclerites characters and size. The detailed morphological feature of *Melithaea delicata* is described below with all other reported 14 species under this genus from Indian coast.

Phylum: Cnidaria Verrill, 1865

Class: Anthozoa Ehrenberg, 1834

Subclass: Octocorallia Haeckel, 1866

Order: Alcyonacea Lamouroux, 1812

Suborder: Scleraxonia Studer, 1887

Family: Melithaeidae Gray, 1870
Melitodidae

Subfamily: Melithaeinae, Alderslade, 2006

Genus: *Melithaea* Milne Edwards, 1857

1857 *Mopsella* Gray

1859 *Acabaria* Gray

1859 *Clathraria* Gray

1870 *Wrightella* Gray

1902 *Pleurocoralloides* Moroff

1. *Melithaea delicata* (Hickson, 1940) (Figs. 1 & 2)
- 1940 *Acabaria delicata* Hickson,
- 1940 *Acabaria gracillima* Stiasny
- 1996 *Acabaria delicata* Weinberg
- 1966 *Acabaria biserialis* Tixier-Durivault
- 1976 *Acabaria biserialis* Grasshoff
- 1987 *Acabaria* aff. *delicata* Ofwegen

Material examined: The specimen deposited in ZSI/ANRC-13682. Live colony and polyps are greenish yellow in colour and internodes are red in colour, the preserved sample was pale yellow with pink colour on internodes; 23 mm length and 30 mm width. The sample was collected from 18 m depth at South Bay, Little Andaman (Lat. 10°30.605 N; Long. 092°29.143 E), on 18 November 2015.

Description: The sample is small, uneven branches and 1.0 mm cylindrical stem. The branches are free, anastomoses and dichotomous forms. Polyps are scattered on the branches and calyces are less than 1.0 mm diameter. The calyx wall is projected above the coenenchyme and pale yellow colour sclerites are noted on the coenenchyme and the calyx. The red colour sclerites are presented in the nodes and internodes (2.0 mm). The coenenchyme have spindle, simple irregular tuberculation sclerites and club shape sclerites presented in the calyx. The nodes and internodes have a rod- and cigar-shaped sclerites.

Sclerites: These are exceedingly varied in shape. The average surface of calyx is 0.10 mm to 1.04 mm clubs and spindles sclerites; the coenenchyme surface have 0.10 mm to 0.175 mm spindles sclerites and 0.05 mm to 0.08 mm clubs and rods sclerites on the nodes surface and subsurface. In collaret, long pointed and small spindle-shaped sclerites are present (0.10 mm – 0.20 mm).

Colour: The living colony is greenish yellow in colour; the tentacles of polyp also in same colour and nodes are red in colour. The coenenchymal sclerites are pale yellow in colour and coenenchymal on nodes are red in colour, nodes surface are red in colour and subsurface of internodes are pale yellow or colourless sclerites.

Distribution: Indonesian; Red Sea; Ghardaqa; Madagascar; Eilat; Kenia; India, Little Andaman, Andaman and Nicobar.

References: Hickson^{27&9}

Remarks: New distribution record from Indian water.

2. *Melithaea andamanensis* (van Ofwegen, 1987)
 1909 *Melithaea variabilis* Thomson and Simpson
 1905 *Melithaea variabilis* Hickson
 1987 *Acabaria andamannensis* van ofwegen

Distribution: Central Indo-Pacific; India, Mandapam, Gulf of Mannar, Cuddalore, Southeast coast of India, South Andaman.

References: Fernando^{20&28}.

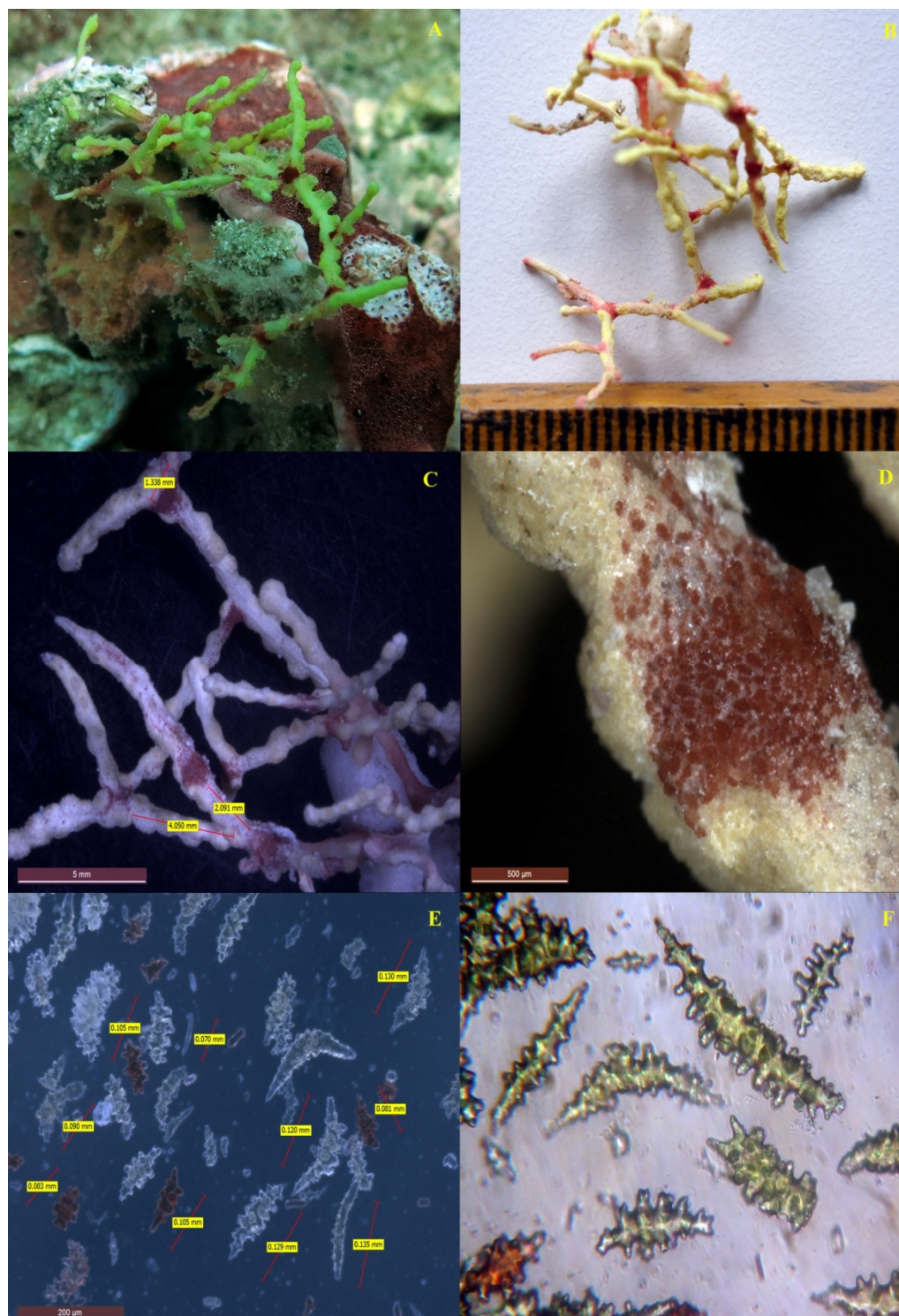


Fig. 1 — *Melithaea delicata*, A. Underwater examination, B. Registered specimen, C. Branching type, D. Node examination (Leica – DFC 500), E-F. Sclerites examination (Labovision AXR 20; Leica – DFC 500).

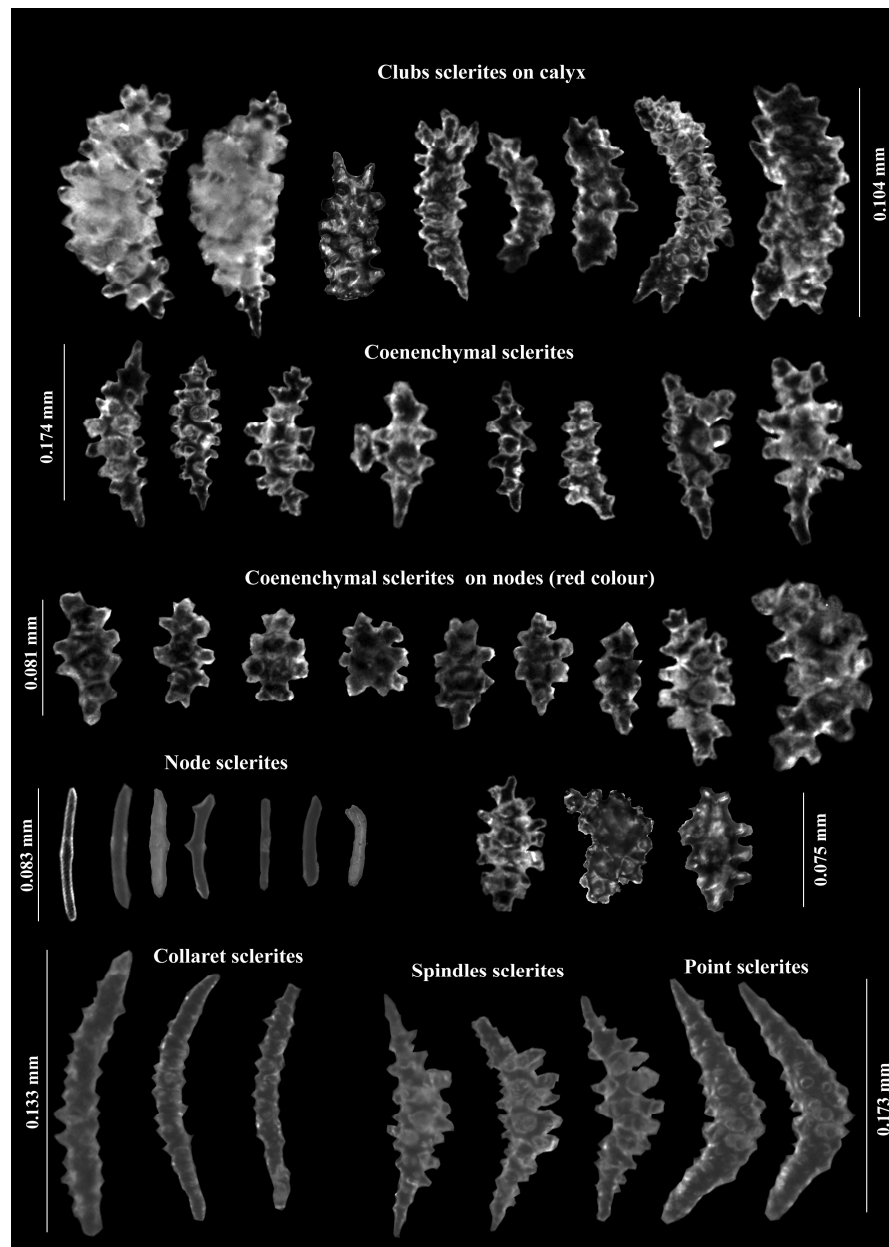


Fig. 2 — Sclerites of *Melithaea delicata*, newly recorded species from Little Andaman, India.

3. *Melithaea biserialis* (Kukenthal, 1908)

1908 *Acabaria biserialis* Kukenthal

1976 *Acabaria biserialis* Grasshoff

1987 *Acabaria biserialis* Ofwegen

2014 *Melithaea biserialis* Reijnen

Distribution: Red Sea; India, Mandapam, Gulf of Mannar, Cuddalore, Southeast coast of India, South Andaman.

References: Kukenthal^{29,20,28}.

4. *Melithaea braueri* (Kukenthal, 1919)

1919 *Wrightella braueri* Kukenthal,

Distribution: Karthip, Nancowry in Andaman and Nicobar Islands.

References: Kukenthal^{30,23,28}.

5. *Melithaea caledonica* Grasshoff, 1999

1970 *Melithaea stormii* Tixier-Durivault

Distribution: Western South Pacific; India, Pongibalu, Munak Gate, Nancowry in Andaman and Nicobar Islands.

References: Grasshoff^{14, 23,28}.

6. *Melithaea cinquemiglia* (Grasshoff, 1999)

1999 *Acabaria cinquemiglia* Grasshoff

- 2014 *Melithaea cinquemiglia* Reijnen
Distribution: Gulf of Mannar, Southeast coast of India, Pongibalu, Andaman and Nicobar Islands.
References: Grasshoff^{14,20,23,28}.
7. *Melithaea hicksoni* (Nutting 1911)
 1911 *Acabaria hicksoni* Nutting
Scirpearia hicksoni Nutting
Distribution: Andaman and Nicobar Islands.
References: Nutting³¹.
8. *Melithaea maldivensis* (van Ofwegen, 1987)
 1987 *Clathraria maldivensis* van Ofwegen,
Distribution: Indian Ocean; India, Lakshadweep.
References: Ofwegen and Vennam³².
9. *Melithaea ochracea* (Linnaeus, 1785)
 1758 *Isis ocracea* Linnaeus
 1767 *Isis ochracea* Linnaeus
 1924 *Melitodes ochracea* Kukenthal
 1970 *Melithaea ochracea* Tixier – Durivault
 1970 *Melithaea rubeola* Tixier – Durivault
 1977 *Melithaea ochracea* Muzik and Wainwright
 1984 *Melithaea ocracea* Zou and Chen
 1987 *Melithaea ochracea* van ofwegen
 1996 *Melithaea ochracea* Goh and Chou
Distribution: Havelock, Andaman and Nicobar Islands.
References: Kumar *et al.*,^{23,28}.
10. *Melithaea ornata* (Thomson and Simpson, 1909)
Distribution: Andaman and Nicobar Islands.
References: Thomson and Simpson¹⁹
11. *Melithaea ouvea* (Grasshoff, 1999)
 1999 *Acabaria ouvea* Grasshoff
 2014 *Melithaea ouvea* Reijnen
Distribution: South Pacific Ocean; India, Gulf of Mannar, Gutor Island, Andaman and Nicobar Islands.
References: Grasshoff^{14,20,23,28}.
12. *Melithaea philippinensis* (Wright and Studer, 1889)
 1889 *Acabaria philippinensis* (Wright & Studer)
Distribution: Andaman and Nicobar Islands.
References: Wright and Studer³³.
13. *Melithaea rubeola* (Wright & Studer, 1889)
 1889 *Acabaria rubeola* Wright & Studer
- 1889 *Mopsella rubeola* Wright & Studer
 1937 *Mopsella rubeola* Hickson
 1996 *Mopsella rubeola* Goh and Chou
 2000 *Mopsella rubeola* Van Ofwegen
 2014 *Melithaea rubeola* Reijnen
Distribution: Yerwadi, Gulf of Mannar, John Lawrence, Andaman and Nicobar Islands.
References: Wright and studer^{33,20,23,28}.
14. *Melithaea squamata* (Nutting, 1911)
 1911 *Melitodes squamata* Nutting
 1924 *Melitodes squamata* Kukenthal
 1956 *Melitodes squamata* Utinomi
 1984 *Melitodes squamata* Ofwegen
 1987 *Melitodes squamata* Ofwegen
Distribution: Red Sea; India, Uvari, Kanyakumari, Tuticorin in Gulf of Mannar, Cuddalore, South-east coast of India.
References: Nutting^{31,23,28}.
15. *Melithaea variabilis* (Hickson, 1905)
 1905 *Acabaria variabilis* (Hickson)
 1905 *Melitodes variabilis* Hickson
Distribution: Landfall Island in Andaman and Nicobar Islands and Laccadive Islands.
References: Hickson^{34,24}.

Discussion

The diversity and distribution of octocorals were explored during the study period from June 2013 to May 2016 in Andaman and Nicobar Islands. Based on these studies, as one new distribution reports, the species *Melithaea delicata* was identified based on the morphological variations between the other species and the sample observed at 18 m depth from Little Andaman Island. The genus *Melithaea* was synonymised from the following genera *Acabaria*, *Clathraria*, *Melithaea*, *Mopsella*, *Wrightella* and the species *M. delicata* was misidentified and synonymised in the name of *Acabaria delicata*, *Acabaria gracillima*, *Acabaria delicata*, *Acabaria biserialis*, *Acabaria biserialis*, and *Acabaria aff. delicata*^{9,17}.

In India, total 15 species (*Melithaea andamanensis*, *M. biserialis*, *M. braueri*, *M. caledonica*, *M. cinquemiglia*, *M. delicata*, *M. hicksoni*, *M. maldivensis*, *M. ochracea*, *M. ornata*, *M. ouvea*, *M. philippinensis*, *M. rubeola*, *M. squamata*, *M. variabilis*) were reported under this genus¹⁸⁻²⁴, among them *M. delicata* was reported first time from Indian water. *M. delicata* nodes are red in colour, branching and

colour of the sample is very similar to *Melithaea variabilis*. The species growth form and sclerites size varied between the other reported species (121 sp.) under this genus from elsewhere's^{6,7,9,16,35}. The present study indicates the addition to the diversity index of octocoral fauna in Indian coast.

Conclusion

The octocorals of India have not been sufficiently studied; it can be anticipated that the *Melithaea* diversity is also quite high. Regrettably, there is also a scarcity of information on *Melithaea* diversity of Indian reef regions, such as those of Gulf of Mannar, Gulf of Kutch, Lakshadweep, and Andaman and Nicobar Islands. At this stage, therefore, it can be cautiously concluded that the number of *Melithaea* species on the Indian reef is rather moderate. It is anticipated that several extensive surveys and taxonomical studies are required to inventorise the octocoral fauna to enhance the database of Andaman and Nicobar Islands as well as mainland reef ecosystem of India.

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